Slide 4: Strategy Summary & Visualizations

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Hierarchical Architecture Overview

1. 1D Row Decomposition

A strict SPMD approach where the Matrix A workload is statically sliced horizontally across completely independent parallel worker nodes.

2. The Core Hypothesis

HPC networks are volatile. I hypothesized that burning local CPU cycles to generate data would bypass cluster network bottlenecks completely.

3. Memory Management

Avoided pointer arrays. All matrices strictly use 1D Contiguous Allocation (`malloc(rows * cols)`) to prevent memory fragmentation.

4. Perfect Load Balancing

Used deterministic offsets to cleanly distribute remainder rows to lower ranks. Maximum imbalance between any two nodes is strictly 1 row.

5. Cache Optimization

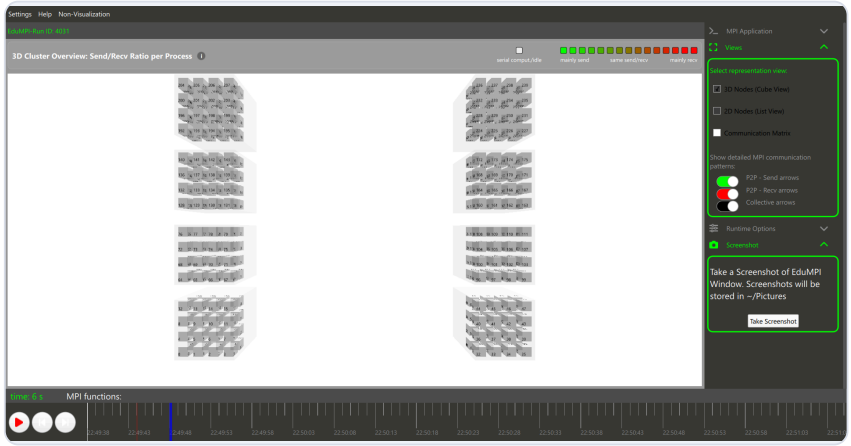
Refactored the compute loop to an `i-k-j` order. This enables linear stride-1 fetching of Matrix B, maximizing L1/L2 cache hits.

6. Zero-Comm Trade-off

Achieved 0% network usage during setup by trading bandwidth for redundant `O(N²)` local PRNG generation of Matrix B on every node.

EduMPI Visualizations

Grey State: Isolated computation with 0% network utilization



Clean Gather: A single MPI_Gatherv operation at program exit

